

E pluribus : A Multi-Task Vision-Query2Contextual Framework for Ecological Research Assistant

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Abstract

E pluribus is a lightweight, multi-task ecological research-assistance framework that integrates visual perception, species classification, image captioning, hybrid literature retrieval, contextual reranking, and local LLM-based synthesis. E-ConvNeXt serves as the shared visual backbone, demonstrating competitive accuracy with reduced parameters and computation. The retrieval pipeline combines BM25, semantic embeddings, taxonomic context, and Cross-Encoder reranking to identify research-relevant evidence from multiple ecological and scholarly sources. A structured information flow further separates observation, inference, evidence, and generation. Preliminary whole-framework evaluation indicates moderate but meaningful ecological assistance capability, while revealing future needs in domain adaptation, retrieval validation, multimodal model integration, and systematic ablation studies.

Keywords: Ecological Research Assistance; E-ConvNeXt; Hybrid Information Retrieval; Context-Aware Re-ranking

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I. Introduction

1.1 Purpose

Ecology and biodiversity science are entering a data-intensive era in which camera traps, autonomous sensors, unmanned aerial vehicles, and digital field-recording systems continuously generate large volumes of visual and environmental observations. However, the scientific value of these data is not determined by collection scale alone. A substantial gap [1] remains between acquiring ecological observations and transforming them into interpretable, traceable, and research-relevant knowledge. Manual inspection, species annotation, literature discovery, evidence comparison, and research-note consolidation remain fragmented and labor-intensive activities. Consequently, the principal bottleneck of modern ecological informatics is increasingly shifting from data acquisition to **data-to-knowledge transformation**.

Existing artificial intelligence applications have partially addressed this problem through object detection, image classification, and automated species recognition. Nevertheless, most systems remain task-specific perception tools. They can identify what may appear in an image, but they rarely connect that prediction to the wider scientific context required by researchers: taxonomic alternatives, habitat and behavioral evidence, related ecological studies, conservation implications, uncertainty, and locally accumulated research records. High classification accuracy alone therefore does not constitute an ecological research assistant. A scientifically useful system must connect perception with contextual retrieval, evidence selection, and knowledge synthesis while preserving the distinction between model prediction and documented evidence.

The purpose of **E pluribus** is to investigate and implement a local, multi-model artificial intelligence framework that converts an ecological image and a researcher’s question into a structured, evidence-oriented research context. The project integrates visual detection, scene interpretation, fine-grained species classification, semantic document retrieval, context-aware reranking, and local large language model synthesis into a unified software workflow. Rather than treating these components as isolated demonstrations, the system is designed as an end-to-end research pipeline in which each module produces an explicit intermediate representation that can be inspected, evaluated, and reused by subsequent modules.

This objective reflects E. O. Wilson’s broader vision of biodiversity science as an expanding and distributed body of knowledge that requires systematic digital integration [2]. Projects such as the Encyclopedia of Life [3] demonstrate the importance of connecting fragmented taxonomic and biological information; E pluribus extends this principle from static knowledge aggregation toward observation-centred knowledge access. A field image is treated not merely as a classification input, but as an entry point into a larger scientific evidence space.

From a modern deep-learning perspective, the project also moves beyond the assumption that pattern recognition alone is sufficient. Inspired by research directions emphasised by Yoshua Bengio [4], the framework separates perception, contextual representation, evidence relevance, and generated interpretation instead of expecting a single model to implicitly solve every reasoning step. In parallel, Karpathy’s Software 2.0 perspective motivates the treatment of datasets, learned representations, evaluation procedures, and model interfaces as first-class software components. Accordingly, the project emphasises modularity,

traceability, uncertainty disclosure, reproducible evaluation, and compatibility with limited local computing resources.

The ultimate purpose is not to replace ecological expertise or automate scientific judgement. It is to reduce the mechanical cost of moving between observation, identification, literature search, evidence screening, and preliminary synthesis. By transforming fragmented research operations into a coherent and inspectable AI-assisted workflow, E pluribus aims to help researchers reach relevant evidence more efficiently, formulate better-informed questions, and devote greater attention to ecological interpretation and scientific discovery.

1.2 Multi-Task Modules Concept

E pluribus adopts a six-module architecture to decompose ecological research assistance into interpretable and independently evaluable stages. M1 detects ecological objects, M2 extracts environmental and scene context, and M3 performs fine-grained species classification. M4 retrieves candidate literature through semantic and metadata-based matching, while M5 applies context-aware reranking to distinguish scientifically relevant evidence from documents that are only superficially similar. M6 integrates the selected literature with local research records and generates evidence-oriented responses. This modular design is necessary because ecological interpretation cannot be reduced to a single prediction task. Separating perception, retrieval, relevance assessment, and synthesis improves traceability, reduces error propagation, supports component-level evaluation, and enables each model to be upgraded without redesigning the complete system.

II. Theoretical Deduction

2.1 E-ConvNeXt

ConvNeXt [6] is a pure convolutional neural network that modernizes conventional CNN design by incorporating several architectural principles associated with Vision Transformers, including large receptive fields, hierarchical feature processing, inverted bottlenecks, and simplified stage organization. E-ConvNeXt [7] extends this foundation with lightweight channel attention and cross-stage partial computation, aiming to retain strong feature representation while substantially reducing model complexity.

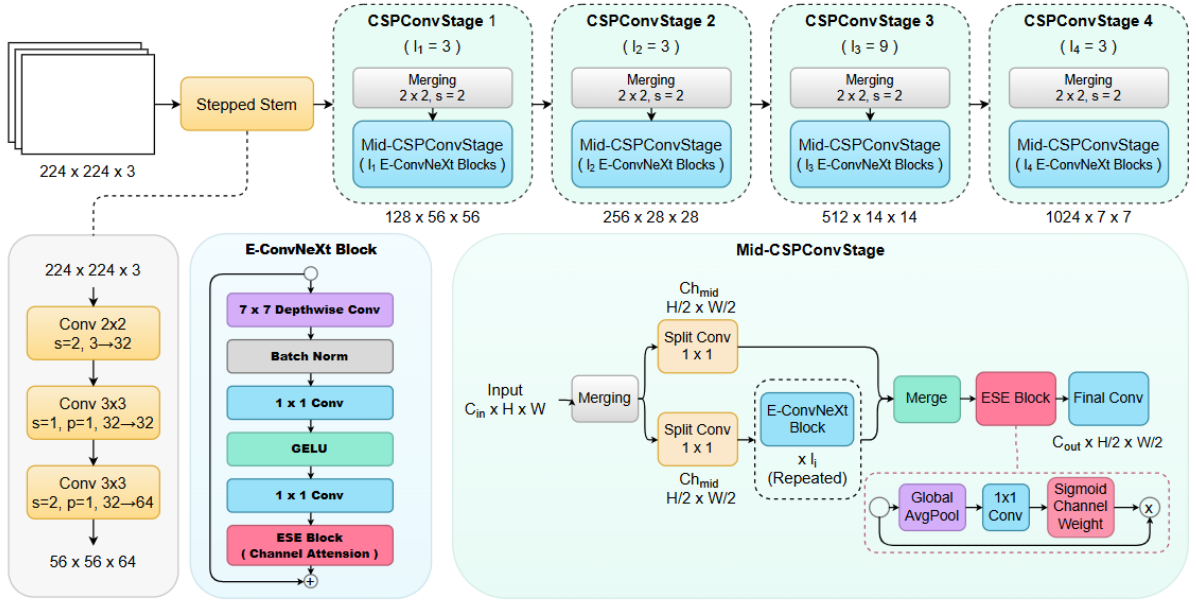


Fig. 1. Architecture diagram of E-ConvNeXt

As illustrated in **Fig. 1**, E-ConvNeXt is organized as a hierarchical visual backbone composed of a Stepped Stem followed by four CSPConvStages. Each stage progressively reduces spatial resolution and expands channel capacity, allowing the network to transform low-level visual patterns into increasingly abstract semantic features. Its architectural contribution is not merely a reduction in parameter count. More importantly, it reorganizes how visual information is

sampled, propagated, divided, transformed, and recalibrated throughout the network.

2.1.1 Stepped Stem

The first architectural component is the **Stepped Stem**, which addresses information loss during early-stage downsampling. The original ConvNeXt stem directly applies a (4×4) convolution with stride 4, effectively converting the input image into non-overlapping patches. Although computationally efficient, such aggressive downsampling may discard fine spatial structures before they are sufficiently encoded. This limitation is particularly relevant to ecological imagery, where species distinctions may depend on small-scale characteristics such as fur texture, ear shape, body contour, or localized markings.

E-ConvNeXt replaces this abrupt operation with a progressive sequence consisting of one (2×2) stride-2 convolution followed by two (3×3) convolutions. The channel dimension is also expanded gradually rather than changed in a single step. The initial convolution preserves the patchification principle of ConvNeXt, while the subsequent convolutions perform additional local feature extraction before the next reduction in spatial resolution. This stepped transformation creates a more controlled transition from raw pixels to latent features, thereby preserving local boundaries and texture information with only a moderate computational increase. Experimental results in the original study show that the Stepped Stem improves classification accuracy without increasing FLOPs relative to the alternative stem configuration.

2.1.2 CSPConvStage

The second key component is the **CSPConvStage**, which introduces the Cross Stage Partial Network principle into ConvNeXt. At the beginning of each stage,

the feature representation is divided into two branches. One branch is processed by a sequence of E-ConvNeXt blocks, while the other retains a shorter transformation path. The two branches are subsequently merged to form the stage output.

This partial-processing strategy improves computational efficiency because only a proportion of the feature channels passes through the complete nonlinear transformation path. At the same time, the bypass branch preserves relatively unmodified information and provides a direct route for gradient propagation. Consequently, the architecture avoids repeatedly applying expensive operations to the entire feature tensor while maintaining both transformed semantic features and lower-distortion information.

E-ConvNeXt further replaces a simple channel-splitting operation with a learnable (1×1) convolution and introduces a transition hyperparameter to control the intermediate channel width. This design permits the network to learn how information should be allocated between branches rather than relying on a fixed split. The stage-depth and channel ratios are then adjusted to recover representational capacity after complexity reduction. Therefore, CSPConvStage should be interpreted as a structured information-allocation mechanism: computation is concentrated on the feature subset that requires deeper transformation, while complementary information is retained through a lightweight path.

2.1.3 E-ConvNeXt Block

The **E-ConvNeXt Block** performs the primary feature transformation within each CSP branch. It begins with a (7×7) depthwise convolution, which enlarges the receptive field and captures broader spatial context at substantially lower cost than

a standard convolution. Two (1×1) convolutions then perform channel expansion and projection, forming an inverted residual structure. This separation between spatial processing and channel mixing allows the network to model local-to-regional visual relationships without applying a computationally expensive dense convolution across all channels.

For greater hardware efficiency, E-ConvNeXt replaces Layer Normalization and fully connected transformations with Batch Normalization and (1×1) convolutions. This removes repeated tensor transposition between channel-first and channel-last layouts and makes the block more compatible with conventional convolutional execution pipelines. The reported ablation study shows that the BN-based block improves throughput while preserving the same FLOPs and parameter scale.

The block also replaces ConvNeXt’s Layer Scale mechanism with an **Effective Squeeze-and-Excitation block**. ESE performs lightweight channel-wise recalibration by summarizing global spatial responses and assigning adaptive importance to individual channels. Informative responses can therefore be strengthened, while redundant or weakly relevant channels are suppressed. This adds attention-like selectivity without introducing the quadratic computation associated with self-attention.

2.1.4 Overall

Overall, E-ConvNeXt combines progressive early feature extraction, partial cross-stage computation, large-kernel depthwise convolution, efficient channel mixing, and lightweight channel attention. The original study reports that E-ConvNeXt-mini achieves 78.3% ImageNet-1K Top-1 accuracy at 0.9 GFLOPs, while E-ConvNeXt-small reaches 81.9% at 3.1 GFLOPs. The architecture also

demonstrates transferability to object detection tasks. These results indicate that its efficiency is derived not from indiscriminate model compression, but from more selective use of computation and more effective preservation of visual information.

E-ConvNeXt was therefore selected as the shared visual backbone of E pluribus because it provides an appropriate balance among fine-grained feature preservation, computational efficiency, model scalability, and downstream-task adaptability.

2.1.5 Supplementary Concepts

Depthwise Convolution.[8] Depthwise convolution applies one spatial filter independently to each input channel rather than jointly processing all channels. It substantially reduces parameters and FLOPs while preserving spatial feature extraction. Channel interaction is subsequently performed using (1×1) pointwise convolutions.

ESE Block.[9] The Effective Squeeze-and-Excitation block aggregates global spatial information and generates channel-wise importance weights through a lightweight transformation. It adaptively emphasizes discriminative channels and suppresses less informative responses, improving feature selectivity with minimal computational and parameter overhead.

2.2 E-ConvNeXt as Backbone of YOLO

As illustrated in **Fig. 2**, the proposed detector replaces the conventional YOLO backbone with E-ConvNeXt while retaining a lightweight detection neck and head derived from YOLOv10n [10]. The objective is to combine the information-efficient visual representation of E-ConvNeXt with the low-latency, end-to-end detection design of modern YOLO architectures. Instead of using only the final

backbone output, feature maps are extracted from three CSPConvStages at progressively reduced spatial resolutions. These outputs provide complementary levels of representation: the high-resolution feature map preserves boundaries and small-object details, the intermediate map captures object parts and regional patterns, and the low-resolution map contains stronger semantic abstraction and a wider effective receptive field.

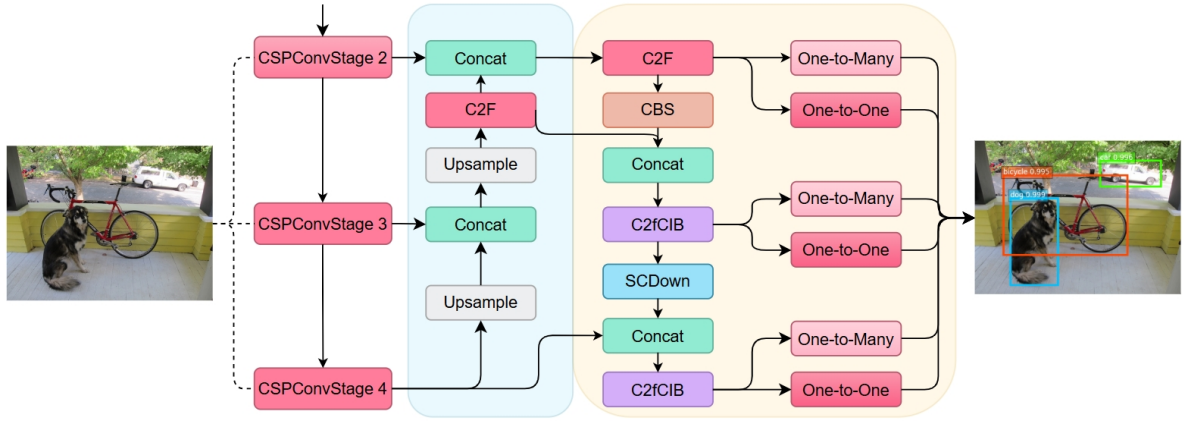


Fig. 2. *E-ConvNeXt PAN-FPN-style YOLO Object Detection Model*

The three feature scales are integrated through a **PAN-FPN-style neck**, rather than a weighted BiFPN. The top-down FPN pathway upsamples deep semantic features and concatenates them with shallower backbone outputs, allowing high-level category information to enrich spatially detailed representations. A subsequent bottom-up path aggregation process transfers the fused localization information back toward lower-resolution levels. This bidirectional flow reduces the semantic gap between backbone stages and produces detection features suitable for objects of different sizes. Unlike BiFPN, however, the present design uses direct concatenation rather than learnable weighted feature fusion.

The neck and detection pathway retain several efficiency-oriented YOLO modules. **C2f** provides cross-stage feature aggregation through multiple lightweight bottleneck paths, improving gradient flow and feature reuse without

requiring every channel to undergo the same transformation depth. **CBS**, consisting of convolution, batch normalization, and a nonlinear activation, performs standardized feature projection, channel alignment, and local refinement between fusion operations. **C2fCIB** incorporates Compact Inverted Blocks into the C2f structure, combining partial feature reuse with computationally economical spatial and channel transformations. It is positioned in deeper stages where semantic processing is important but spatial dimensions are smaller. **SCDown** applies spatial–channel decoupled downsampling: spatial resolution is reduced using an efficient depthwise operation, while channel projection is handled separately through pointwise convolution. This avoids the higher computational cost of simultaneously performing dense spatial and channel transformation in a conventional stride convolution.

Each output scale is connected to two parallel detection branches based on YOLOv10’s consistent dual-assignment strategy. During training, the **one-to-many head** assigns multiple positive predictions to each ground-truth instance, providing dense supervision, stronger gradients, and faster convergence. However, such predictions are inherently redundant and would ordinarily require non-maximum suppression to remove overlapping boxes. The **one-to-one head** instead learns a unique correspondence between each object and a single prediction. Both heads share a consistent matching metric so that the rich supervision learned from one-to-many assignment supports the optimization of the one-to-one branch. During inference, only the one-to-one predictions are retained. Because each object is represented by a unique high-confidence prediction, redundant detections are suppressed by the learned assignment mechanism itself, eliminating the latency and threshold sensitivity introduced by conventional NMS post-processing.

2.3 Light Weight Tiny Image Captioning Model

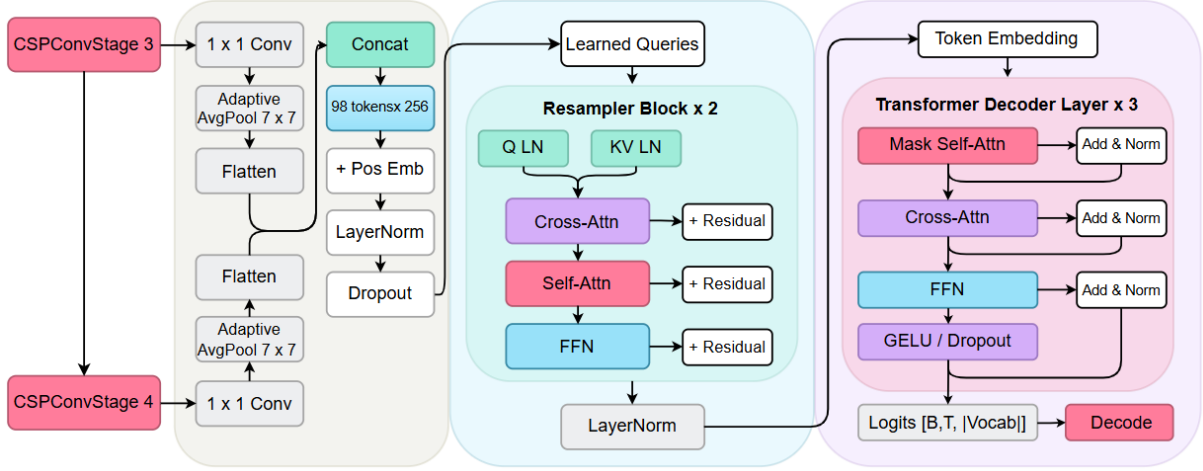


Fig. 3. Architecture Diagram of E-ConvNeXt-Transformer

As illustrated in **Fig. 3**, the proposed Tiny Image Captioning Model uses the third and fourth CSPConvStages of E-ConvNeXt as two complementary visual inputs. The Stage 3 feature map retains comparatively higher spatial resolution and therefore preserves object boundaries, local structures, and scene details, whereas the Stage 4 output provides stronger semantic abstraction and a broader receptive field. Both feature maps are first projected through (1 x 1) convolutions to align their channel dimensions, followed by adaptive average pooling to a fixed (7 x 7) resolution. After flattening and concatenation, the two scales form 98 visual tokens, which are combined with positional embeddings, normalized, and regularized before entering the captioning module [11].

To reduce the computational cost of directly processing all visual tokens, the architecture introduces a two-layer **resampler block** driven by a compact set of learned queries. Through cross-attention, these queries selectively extract the most informative visual evidence from the multi-scale token sequence. Subsequent self-attention and feed-forward transformations model relationships among the compressed latent representations. This mechanism produces a smaller

and more task-specific visual memory, reducing attention complexity while retaining relevant object and environmental context.

Caption generation is performed by a lightweight three-layer Transformer decoder. Masked self-attention models dependencies among previously generated words, while cross-attention connects the textual sequence to the resampled visual representation. The feed-forward network, GELU activation, residual connections, normalization, and dropout provide nonlinear transformation and stable optimization. Finally, token logits are projected over the vocabulary and decoded autoregressively.

The model is designed to generate concise scene-level descriptions rather than highly detailed natural-language narratives. Its primary role is to convert E-ConvNeXt features into compact semantic context that can support downstream query expansion, literature retrieval, and ecological interpretation under limited computational resources.

2.4 Hybrid Embedding Retrieval Framework

As illustrated in **Fig. 4**, the Hybrid Embedding Retrieval Framework is designed to transform a user’s ecological question and the species prediction produced by M3 into a structured, high-recall set of candidate documents. The framework does not treat the original query as an isolated text string. Instead, it constructs a taxonomy-aware query pack containing the scientific name, genus, common names, and semantically related query variants. Taxonomic context is supplemented through external biodiversity resources such as GBIF, iNaturalist, and Animal Diversity Web, allowing the retrieval process to preserve species identity even when different literature sources use alternative names, abbreviations, or classification levels.

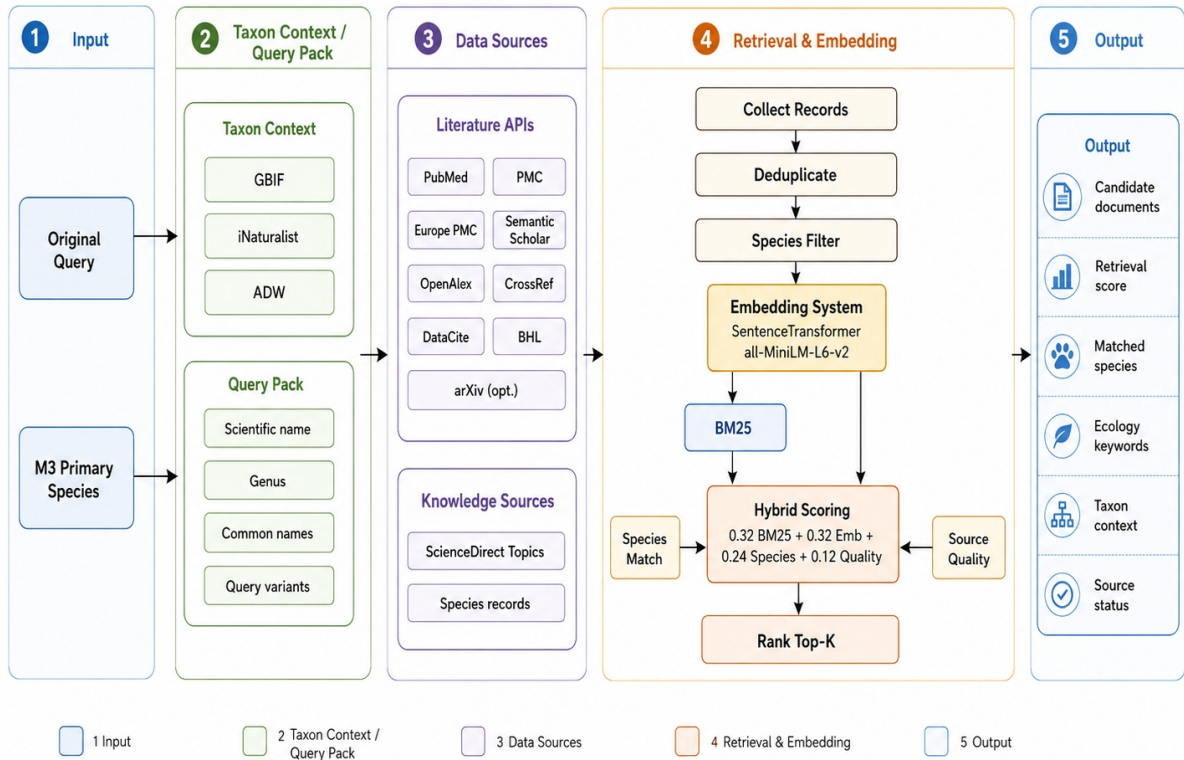


Fig. 4. Framework diagram of hybrid retrieval

Candidate records are collected from heterogeneous scholarly services, including PubMed, PubMed Central, Europe PMC, Semantic Scholar, OpenAlex, Crossref, DataCite, and the Biodiversity Heritage Library. Because these sources differ in metadata completeness, document structure, and identifier conventions, the retrieved records are normalized into a common schema containing titles, abstracts, keywords, publication years, persistent identifiers, and source information. DOI-, title-, and metadata-based deduplication is then applied to prevent repeated publications from dominating the ranking. A species filter removes records that are clearly inconsistent with the predicted taxon while retaining broader genus-level or ecologically relevant studies when exact species mentions are unavailable.

The retrieval stage combines **sparse lexical matching** and **dense semantic representation**. BM25 evaluates explicit term correspondence using term

frequency, inverse document frequency, and document-length normalization [12]. It is therefore effective for scientific names, habitat terminology, behavioral keywords, and other exact expressions whose lexical form is important. In parallel, the SentenceTransformer model all-MiniLM-L6-v2 encodes the query and document metadata into dense vectors. Following the Sentence-BERT retrieval paradigm, semantic similarity can be estimated efficiently through cosine similarity between independently encoded representations [13]. This dense component captures conceptual relationships that may not share identical vocabulary, such as links between “urban adaptation” and descriptions of anthropogenic habitat use.

The two retrieval signals are combined with domain-specific evidence through a normalized hybrid score:

$$S(d, q) = 0.32 S_{BM25} + 0.32 S_{emb} + 0.24 S_{species} + 0.12 S_{quality} \quad (1)$$

Here, $S_{species}$ measures consistency between the document and the taxonomic context, while $S_{quality}$ reflects metadata completeness, source reliability, and document accessibility. The coefficients are initial engineering priors rather than universally optimal constants and should subsequently be calibrated using retrieval metrics such as Recall@K, Precision@K, Mean Reciprocal Rank, and nDCG. This combination is motivated by the complementary behavior of sparse and dense retrieval: lexical models preserve exact scientific terminology, whereas neural embeddings recover semantically related evidence beyond direct word overlap [14].

The final output consists of the ranked Top-(K) candidate documents together with retrieval scores, matched species, ecological keywords, taxonomic context, and source status. These structured outputs preserve provenance and expose the

factors behind document selection, while supplying a broad but controlled candidate set for the subsequent context-aware reranking module.

2.5 Contextual Cross Encoder Re-ranker

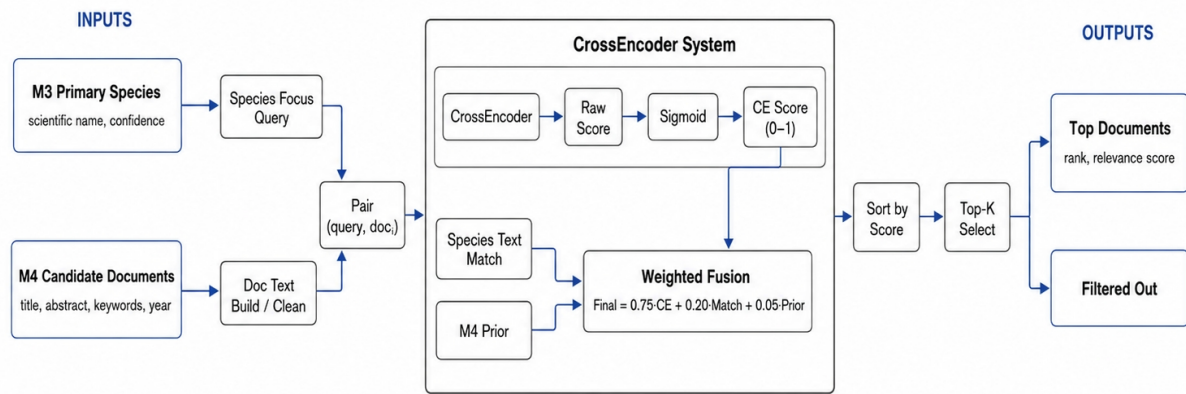


Fig. 5. Framework of cross encoder re-ranker

The Contextual Cross-Encoder Re-ranker forms the second-stage precision layer of the literature retrieval pipeline. While the hybrid retrieval framework is designed to retrieve a broad set of potentially relevant documents, lexical and embedding similarities alone cannot reliably determine whether a publication matches the complete ecological intent of a query. A document may mention the correct species but focus on an unrelated biological mechanism, or share habitat terminology without addressing the behavior, conservation issue, or observational context required by the researcher. The re-ranker is therefore introduced to distinguish genuine contextual relevance from superficial textual similarity.

For each candidate document, the model jointly encodes the research query, taxonomic information, optional visual context, and document metadata as a single paired input. Unlike a bi-encoder, which independently represents the query and document before comparing their vector similarity, a Cross-Encoder allows attention to operate directly across both sequences. This enables token-level interaction between scientific names, ecological terms, behavioral descriptions,

habitat evidence, and the researcher’s intended question. BERT-based passage re-ranking has demonstrated that such joint encoding can substantially improve ranking precision because the model evaluates relevance through contextual interaction rather than fixed embedding distance [15].

In the proposed framework, the Cross-Encoder receives only the Top-(K) documents produced by the hybrid retriever, making its higher computational cost manageable. Each query–document pair is assigned a contextual relevance score, after which the candidates are reordered and the highest-ranked evidence is forwarded to the language model. A compact pretrained encoder, such as MiniLM or DistilBERT, may be used to balance semantic discrimination with local inference efficiency.

The model can be trained using manually reviewed examples or weak relevance labels generated from explicit ecological criteria, including taxonomic consistency, habitat correspondence, behavioral relevance, conservation intent, and source adequacy. This design prevents the downstream language model from receiving a large collection of merely related documents and instead supplies a smaller, more coherent evidence set that better reflects the researcher’s actual information need.

2.6 Design of Information Flow

The information flow of E pluribus is designed to preserve the distinction between observation, model inference, retrieved evidence, and generated interpretation. The user query and field image enter the system through separate but coordinated paths. The image is analyzed by the scene-context and species-classification modules, while the original query directly guides literature retrieval and context construction. This parallel structure prevents the visual prediction from replacing

the researcher’s actual question. Instead, species hypotheses, environmental cues, and behavioral descriptions are treated as supplementary signals that refine retrieval and interpretation. Candidate documents are first collected by the retrieval module and then filtered by the context-aware re-ranker, ensuring that only evidence consistent with both the query and ecological context proceeds to the generation stage.

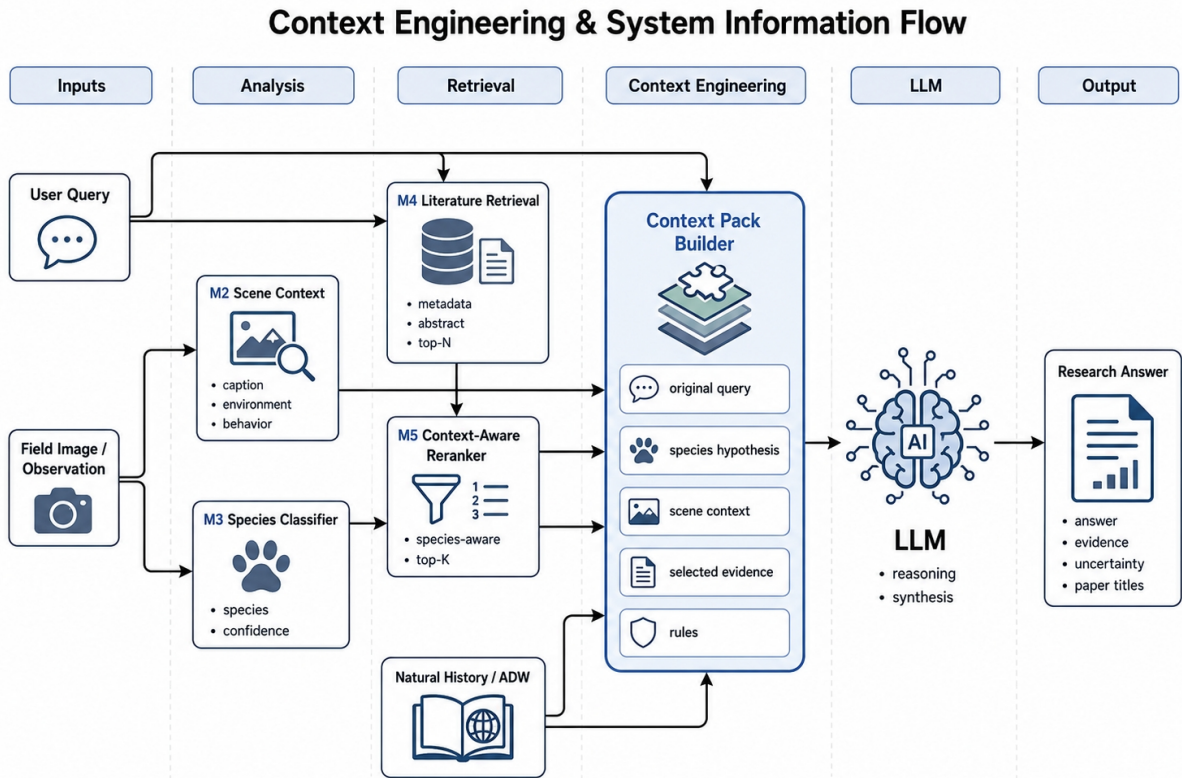


Fig. 6. Context Engineering and System Information Flow of *E pluribus*.

The Context Pack Builder serves as the central information-control layer. It combines the original query, species hypothesis, scene context, selected literature, natural-history knowledge, and response constraints into a structured evidence package before the data are passed to the LLM. This design limits uncontrolled prompt composition and reduces the risk that irrelevant retrieval results or uncertain visual predictions dominate the final response. The LLM therefore

operates primarily as a synthesis component rather than an independent source of facts. By maintaining explicit intermediate outputs and traceable transitions between modules, the system supports uncertainty disclosure, evidence inspection, component-level evaluation, and future replacement of individual models without altering the complete research workflow.

2.7 Design of Feature Engineering & Dataset

The dataset strategy is organized according to the function of each module. Species images are collected through controlled web crawling and metadata verification, while Imagenette may be used for optional pretraining experiments to evaluate transferability. Object detection is initially trained on VOC2007/2012 and subsequently fine-tuned using wildlife-oriented Kaggle datasets to improve ecological-domain adaptation. The image-captioning module uses Flickr30k to learn general visual–language alignment before being applied to ecological scene interpretation. For literature retrieval, the system connects to approximately ten biodiversity databases, scholarly APIs, and natural-history resources.

Feature engineering combines taxonomic normalization, metadata cleaning, class balancing, and strong image augmentation techniques, including random cropping, flipping, color perturbation, scale variation, and geometric transformation. These operations improve robustness to viewpoint, illumination, background complexity, and limited species samples while reducing overfitting across heterogeneous ecological data sources.

III. Experimental Results

3.1 E-ConvNeXt Experimental Results

Metric	ResNet18	E-ConvNeXt-mini	Comparison
 Train Accuracy (%)	99.93%	99.48%	 Higher is better
 Test Accuracy (%)	88.28%	88.76%	 Higher is better
 Epochs	100	100	 Equal
 GFLOPs	7.2543	2.4593	 Lower is better
 Params	11.18M	2.58M	 Lower is better
 Inference Time (ms)	0.4032	0.3784	 Lower is better

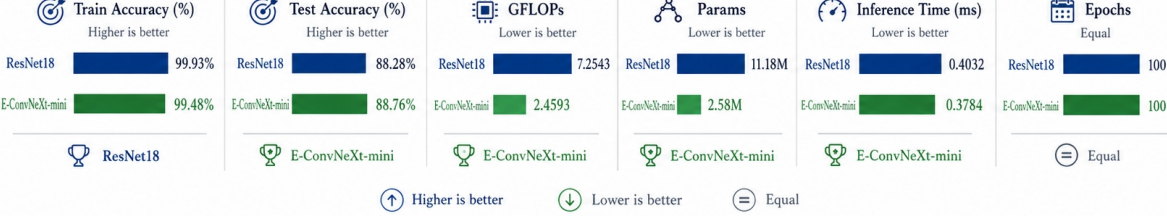


Fig. 7. Experiment result of E-ConvNeXt

A preliminary image-classification experiment was conducted on a laptop equipped with an NVIDIA GeForce RTX 4060 Laptop GPU to evaluate whether E-ConvNeXt-mini provides a practical balance between recognition accuracy and computational efficiency. The dataset consisted of nine fox-related categories collected from heterogeneous web image sources. These categories represented visually similar taxa, including the red fox (*Vulpes vulpes*), Arctic fox (*Vulpes lagopus*), swift fox (*Vulpes velox*), fennec fox (*Vulpes zerda*), and other morphologically related fox species. This task is more demanding than coarse animal classification because several categories share similar body proportions, coat patterns, facial structures, and environmental backgrounds. The model must therefore learn fine-grained discriminative cues rather than relying only on general animal shape.

The E-ConvNeXt-mini training architecture followed a conventional supervised classification pipeline. Input images were resized and normalized before being

processed by the Stepped Stem and hierarchical CSPConvStages. The backbone progressively extracted local texture, body contour, facial morphology, and higher-level species representations. Global average pooling was then applied to the final feature map, followed by a classification layer producing probabilities for the nine categories. Strong image augmentation was used during training to increase variation in viewpoint, scale, illumination, background, and object position. Both E-ConvNeXt-mini and ResNet18 were trained for 100 epochs under the same experimental setting to provide a controlled architectural comparison.

As shown in Fig. 6, ResNet18 achieved a training accuracy of 99.93% and a test accuracy of 88.28%, whereas E-ConvNeXt-mini achieved 99.48% training accuracy and 88.76% test accuracy. Although E-ConvNeXt-mini produced a slightly lower training accuracy, its test accuracy was 0.48 percentage points higher. This result suggests that the proposed architecture learned a representation with marginally better generalization rather than simply fitting the training samples more aggressively. The small difference between the two models should not be interpreted as a definitive accuracy advantage without repeated trials; however, the result is meaningful when considered together with the substantial reduction in computational complexity.

E-ConvNeXt-mini required 2.4593 GFLOPs, compared with 7.2543 GFLOPs for ResNet18, corresponding to an approximate 66.1% reduction in computation. Its parameter count was reduced from 11.18 million to 2.58 million, representing approximately 76.9% fewer parameters. The measured inference time also decreased from 0.4032 ms to 0.3784 ms under the same environment. Although the latency difference is relatively modest on a GPU with high parallel-processing capability, the reduction in parameters and FLOPs is significant for memory-

constrained deployment and for systems in which the backbone is shared by multiple downstream tasks.

These results indicate that E-ConvNeXt-mini is a highly promising visual backbone. Its performance does not result solely from model compression; rather, the Stepped Stem, cross-stage partial feature processing, depthwise convolution, and lightweight channel recalibration provide a more selective allocation of computation. The architecture preserves sufficient fine-grained information for visually similar fox categories while avoiding unnecessary full-channel transformation at every stage. Under an appropriate training framework, it therefore maintains competitive classification accuracy with considerably lower computational and parameter costs.

The experiment provides practical evidence that E-ConvNeXt-mini is suitable for the proposed multi-task ecological system, particularly where detection, classification, and image-captioning modules must share limited local computing resources. Nevertheless, future evaluation should include repeated runs, confidence intervals, per-class precision and recall, confusion-matrix analysis, and testing on independently collected field images to verify the robustness of the observed advantage.

3.2 E-ConvNeXt-YOLO Experimental Results

3.2.1 Discussion of E-ConvNeXt-YOLO

The E-ConvNeXt-YOLO detector was evaluated through a two-stage training procedure on an NVIDIA RTX 3070 GPU. The model was first pretrained on the PASCAL VOC 2007/2012 datasets to learn general object localization, category discrimination, and bounding-box regression, and was subsequently fine-tuned using a wildlife-oriented Kaggle dataset to adapt the detector to ecological

imagery. YOLOv10 was trained under the same epoch schedule and evaluation configuration as the principal baseline. Both models used the one-to-one prediction branch during evaluation, and the complete test set contained 12,032 ground-truth objects.

As shown in Fig. 7, both models exhibited stable loss reduction throughout the 100 training epochs. E-ConvNeXt-YOLO decreased from an initial training loss of 6.50 to 2.66, whereas YOLOv10 decreased from 7.10 to 3.02. The consistently lower loss curve indicates that the E-ConvNeXt backbone provided more effective feature representations for the shared detection framework. Its progressive stem, cross-stage partial processing, and depthwise spatial modeling appear to support more efficient optimization of localization and classification objectives.

On the complete test set, E-ConvNeXt-YOLO achieved a final mAP of 0.5713, compared with 0.5365 for YOLOv10. The best recorded full-test mAP values were 0.5779 and 0.5382, respectively. E-ConvNeXt-YOLO therefore produced an absolute improvement of approximately 3.5 percentage points over the baseline. The accuracy trend also shows that most performance gains occurred before approximately 60 epochs, after which both models entered a plateau region. E-ConvNeXt-YOLO nevertheless maintained a consistent advantage throughout the later training stage.

These results provide preliminary evidence that E-ConvNeXt is not only effective for image classification but can also serve as a competitive multi-scale detection backbone. Under the same training framework, it achieved lower optimization loss and higher detection accuracy than the reference YOLOv10 configuration. However, the current experiment should be interpreted as architectural validation rather than a complete deployment benchmark, because detection performance

remains dependent on dataset diversity, annotation quality, pretraining domain, and fine-tuning strategy.

3.2.2 Image Domain and Pretraining Effects

The wildlife fine-tuning experiment further demonstrates the importance of domain alignment. VOC pretraining provides useful generic representations of object boundaries and foreground–background structure, but its categories and visual distribution differ substantially from wildlife imagery. Ecological scenes contain greater variation in illumination, vegetation, camouflage, object scale, occlusion, and background complexity. Consequently, a detector pretrained primarily on general-purpose imagery may require stronger domain-specific adaptation before its learned representations become fully effective.

In additional wildlife experiments, an extensively pretrained Ultralytics YOLO12 implementation was able to approach approximately 0.80 mAP more rapidly, whereas E-ConvNeXt-YOLO tended to plateau within the 0.50–0.60 range. This difference does not necessarily imply that the proposed backbone is structurally inferior. The commercial-grade YOLO implementation benefits from mature training recipes, large-scale pretrained weights, optimized augmentation, assignment strategies, loss calibration, and extensively validated hyperparameters. By contrast, E-ConvNeXt-YOLO was trained within a comparatively limited experimental framework and without an equivalent scale of detection-specific pretraining.

The observed plateau therefore highlights a training-system limitation as much as a model limitation. Future improvement should prioritize large-scale detection pretraining, wildlife-specific augmentation, class-balanced sampling, learning-rate and loss-weight calibration, higher-resolution inputs, and systematic ablation

of the neck and detection head. This comparison indicates that backbone innovation alone is insufficient; competitive object detection requires coordinated design across representation learning, domain adaptation, optimization, and data engineering.

3.3 E-ConvNeXt-Light Transformer Image Captioning

The proposed E-ConvNeXt–Light Transformer captioning model was evaluated against a conventional ResNet–Attention LSTM baseline to examine whether a compact Transformer-based decoder could reduce model complexity while preserving acceptable caption-generation performance. The proposed architecture combines multi-scale E-ConvNeXt features with learned-query resampling and a lightweight Transformer decoder. This design compresses the visual representation before autoregressive text generation, thereby limiting the computational burden associated with processing a large number of image tokens.

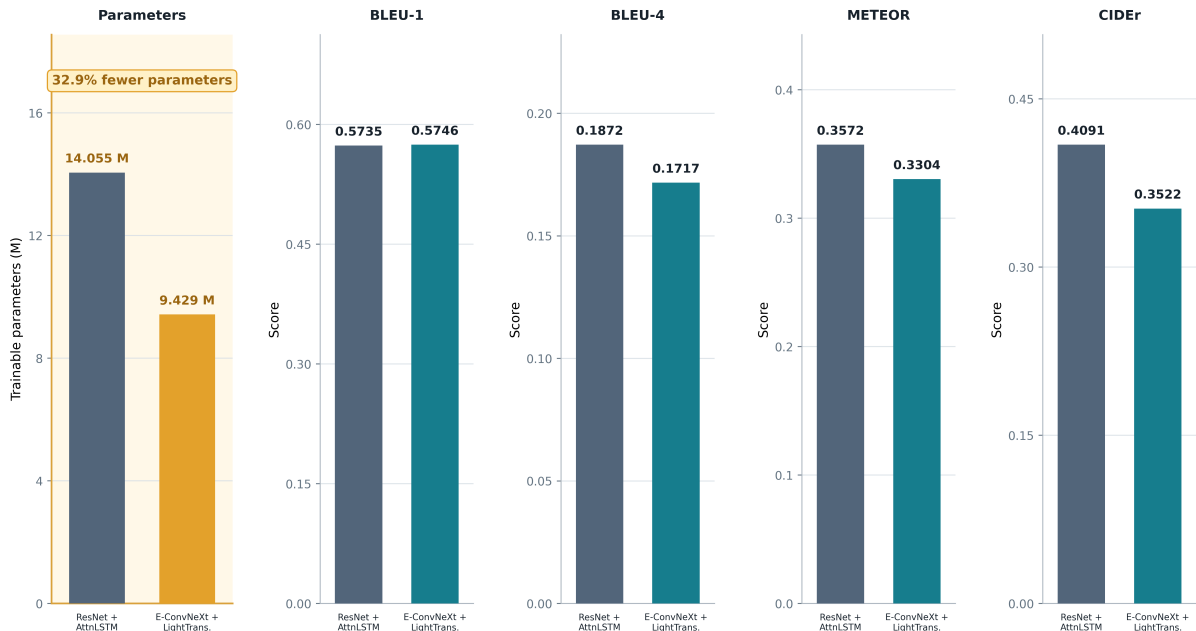


Fig. 8. *image captioning model comparison ratio*

As shown in **Fig. 8**, the baseline model contains 14.055 million trainable parameters, whereas the proposed E-ConvNeXt–Light Transformer requires only

9.429 million. This corresponds to a reduction of approximately 32.9%, demonstrating a substantial improvement in parameter efficiency. Despite the smaller model size, the proposed method achieved a BLEU-1 score of 0.5746, slightly exceeding the baseline score of 0.5735. This indicates that the lightweight model retains comparable capability in generating individual words that match the reference captions.

However, the proposed architecture achieved lower results on higher-order and semantic captioning metrics. Its BLEU-4 score decreased from 0.1872 to 0.1717, METEOR decreased from 0.3572 to 0.3304, and CIDEr decreased from 0.4091 to 0.3522. These results suggest that, although the model preserves basic object and scene vocabulary, its reduced capacity limits the generation of longer phrase structures, detailed sentence composition, and consensus-level semantic descriptions. The largest difference appears in CIDEr, indicating that the generated captions may contain relevant words but do not consistently reproduce the descriptive patterns present across multiple reference captions.

The experiment therefore reveals a clear efficiency–quality trade-off. The proposed method does not outperform the baseline across all linguistic metrics, but it achieves comparable unigram-level recognition with nearly one-third fewer parameters. This makes it suitable for applications in which the caption is used as a compact intermediate representation rather than a final natural-language product. Within E pluribus, the captioning module primarily provides scene, environment, object, and behavioral cues for downstream retrieval and context construction; therefore, concise and computationally efficient descriptions may be more valuable than stylistically complete captions.

It should also be recognized that contemporary image captioning research has increasingly shifted from training isolated captioning models on datasets such as

Flickr30k toward pretrained vision–language models, multimodal foundation models, and instruction-tuned decoders. Future development could therefore retain E-ConvNeXt as the lightweight visual encoder while replacing the current decoder with a pretrained language module, knowledge-distillation framework, or parameter-efficient multimodal adapter. Such extensions may improve semantic richness and ecological terminology without sacrificing the compact deployment characteristics demonstrated in this experiment.

3.4 Contextual Retrieval and Information Flow

The contextual retrieval subsystem demonstrates the practical value of treating ecological information access as a staged evidence-selection problem rather than a single similarity search. The hybrid retriever combines lexical matching, dense semantic similarity, taxonomic consistency, and source-quality signals to construct a broad but controlled candidate set. This division is important because ecological literature contains scientific names, historical synonyms, habitat terminology, and conceptually related expressions that cannot be represented reliably by only one retrieval method. BM25 preserves exact taxonomic and technical terms, whereas embedding retrieval captures semantic relationships beyond direct word overlap. Taxon-aware filtering further reduces false matches caused by common names or closely related species, and source normalization prevents duplicated or incomplete records from disproportionately affecting ranking.

The Cross-Encoder re-ranker provides a second level of discrimination by jointly evaluating the research question, species hypothesis, visual context, and candidate document. Its role is not simply to improve a numerical similarity score, but to determine whether a document satisfies the combined scientific intent of the

observation. For example, a paper may mention the correct species while addressing an unrelated physiological topic, or discuss the correct habitat without containing evidence relevant to the observed behavior. By modeling direct interactions between query and document tokens, the re-ranker can prioritize evidence that is simultaneously consistent with taxonomic identity, ecological context, and research purpose. This reduces the quantity of weakly related material passed to the language model and establishes a clearer boundary between evidence retrieval and answer generation.

The information-flow design strengthens this process by preserving the origin and uncertainty of each signal. The original query, predicted species, scene description, retrieved documents, and natural-history references remain identifiable components within the Context Pack rather than being merged into an unstructured prompt. Consequently, the LLM functions primarily as a controlled synthesis layer. It receives selected evidence and explicit response rules instead of being expected to infer facts independently. This architecture improves traceability, enables inspection of intermediate decisions, and allows errors to be localized to perception, retrieval, reranking, or generation. More fundamentally, it reframes the system from an image-to-answer model into an observation-to-evidence workflow, which is more compatible with scientific practice.

The present implementation nevertheless represents an initial system-level validation. The contribution of each component must be quantified through controlled evaluation. Future experiments will compare BM25-only, embedding-only, hybrid retrieval, and hybrid retrieval with Cross-Encoder reranking using Recall@K, Precision@K, MRR, and nDCG. Ablation studies will separately remove taxonomic expansion, visual context, source-quality scoring, and reranking to determine their individual effects. Expert or rubric-guided relevance

judgments should also be introduced to assess whether the selected documents genuinely support the researcher’s query. These evaluations will clarify whether the framework improves not only retrieval similarity, but also evidence quality, answer grounding, and the reliability of ecological research assistance.

3.5 Whole Framework Discussion

To conduct an initial end-to-end evaluation, GPT-5.5 was used to design ten ecology-oriented research queries covering species identification, habitat selection, behavior, urbanization, diet, seasonal morphology, interspecific competition, road mortality, disease, and subsequent research planning. A single representative wildlife image was then processed through the complete E pluribus pipeline, including visual analysis, literature retrieval, contextual reranking, context-pack construction, and response generation. The final research-oriented answers were produced using Llama as the local LLM and were evaluated according to relevance, ecological usefulness, evidence integration, and response completeness.

The mean score across the ten queries was approximately 5.7/10. The framework performed better on habitat interpretation and research-direction synthesis, but remained weaker in precise species identification, road-ecology reasoning, and health-related assessment. These results do not indicate a highly mature research assistant; however, they demonstrate that the system has moved beyond isolated image classification or generic RAG. The outputs followed the intended ecological-assistance workflow by connecting observations with species hypotheses, contextual literature, and research-oriented interpretation. Future work will strengthen retrieval evaluation, expert annotation, prompt control, and domain-specific LLM adaptation.

IV. Conclusions and Future Directions

E pluribus was developed to investigate whether ecological images, species inference, literature retrieval, contextual evidence selection, and local language-model generation could be integrated into a coherent research-assistance workflow. The central contribution of the project is therefore not any single classifier, detector, captioning model, or retrieval component. Its significance lies in the construction of an interpretable information pipeline that transforms a field observation into a structured evidence context. Instead of treating ecological artificial intelligence as an isolated perception problem, the framework connects visual understanding with scientific information access, document relevance assessment, and evidence-oriented synthesis.

The experiments demonstrate that this direction is technically feasible, but they also reveal a more important conclusion: the quality of a research-assistance system cannot be inferred from the performance of one model alone. E-ConvNeXt-mini showed competitive classification accuracy while requiring substantially fewer parameters and computational operations than ResNet18. E-ConvNeXt-YOLO also achieved lower training loss and higher test mAP than the corresponding YOLOv10 baseline under the same experimental configuration. These results support the hypothesis that efficient information allocation, progressive feature extraction, cross-stage partial computation, and lightweight channel recalibration can provide a useful representation basis for ecological vision tasks.

However, the project also shows that architectural novelty and practical value are not always equivalent. A custom backbone may perform well in controlled experiments, but its final usefulness depends on pretrained weights, optimization maturity, dataset scale, deployment libraries, and compatibility with the

surrounding software ecosystem. This distinction is particularly important for the object-detection module. Training E-ConvNeXt-YOLO from VOC-based initialization provided valuable architectural evidence, but it also exposed the limitations of building a complete detector from comparatively limited data and hardware resources. Modern open-source implementations such as YOLOv11 and YOLOv12 benefit from highly optimized training recipes, large-scale pretraining, mature augmentation strategies, export support, and deployment tooling. In a practical product, adopting and adapting an engineering-ready detector may therefore generate more value than insisting on a fully customized architecture.

This does not eliminate the value of E-ConvNeXt. Rather, it changes how the model should be positioned. E-ConvNeXt should be viewed as one candidate within a wider design space of efficient visual backbones. Future work should compare it with alternative lightweight architectures such as MobileNetV4, FasterNet, GhostNetV2, EdgeNeXt, MambaOut, MobileMamba, and other hardware-aware CNN or state-space designs. The objective should not be to identify a universally superior network, but to determine which architectural principles best match the actual operating conditions of ecological applications. These conditions include small and partially occluded animals, visually similar species, heterogeneous field environments, limited training data, and deployment on laptops or edge devices. Latency, memory consumption, energy usage, export compatibility, and robustness under domain shift should therefore be evaluated together with classification or detection accuracy.

The image-captioning experiment produced an equally important methodological reflection. The proposed E-ConvNeXt-Light Transformer reduced trainable parameters by approximately one-third and preserved competitive BLEU-1 performance, demonstrating a meaningful efficiency advantage. Nevertheless,

lower BLEU-4, METEOR, and CIDEr scores indicate that the compact architecture was less capable of generating complete and semantically rich descriptions. More fundamentally, both the Attention-LSTM baseline and the manually designed lightweight Transformer belong to an earlier captioning paradigm in which visual encoders and language decoders are trained almost independently on caption datasets such as Flickr30k.

This experiment suggests that further effort should not be concentrated solely on optimizing another small captioning decoder within the same paradigm. Contemporary visual-language systems provide stronger semantic priors, broader vocabulary, and more flexible interaction between images and language. Future work should instead investigate how a pretrained vision-language model can be used efficiently rather than retrained comprehensively. Possible directions include freezing a compact visual-language encoder, training only a projection adapter, distilling knowledge from a larger multimodal model, applying low-rank adaptation, or using learned visual tokens to condition a small language decoder. The research question would then shift from “How can a captioning model be trained from the beginning?” to “How can pretrained multimodal knowledge be compressed and controlled for ecological context generation?” This would better align the module with its real function in *E pluribus*: producing concise environmental and behavioral cues for retrieval, rather than generating aesthetically complete captions.

The retrieval subsystem represents one of the most promising parts of the framework because it addresses a problem that remains relevant even when visual models change. The hybrid retrieval design combines exact lexical matching, semantic embeddings, taxonomic consistency, and source-quality signals. This is particularly suitable for ecological literature, where scientific names, synonyms,

historical classifications, habitat terms, and behavioral concepts may appear in inconsistent forms across databases. The current implementation demonstrates how multiple scholarly and biodiversity sources can be normalized into a common candidate-document representation.

Nevertheless, the retrieval coefficients and filtering rules remain primarily engineering assumptions. Future work must replace fixed heuristics with systematically calibrated retrieval strategies. BM25-only, embedding-only, and hybrid configurations should be compared under the same benchmark. Query expansion should be examined separately for scientific names, genus information, common names, habitat descriptors, and visual context. Reciprocal-rank fusion, learned sparse retrieval, domain-adapted scientific encoders, metadata-aware embeddings, and multi-vector document representations may provide stronger alternatives to simple weighted scoring. The evaluation dataset should also include hard negatives: papers that mention the correct species but address an irrelevant question, and papers that omit the exact species name while containing highly relevant ecological evidence.

The Contextual Cross-Encoder is conceptually central but currently insufficiently validated. Its intended role is to determine whether a document is relevant to the full research context rather than merely similar to a query. However, without a carefully labeled ecological relevance dataset, the model's contribution cannot yet be established quantitatively. Future development should construct explicit relevance rubrics covering species consistency, habitat correspondence, behavioral relevance, conservation intent, methodological usefulness, evidence quality, and temporal or geographical applicability.

A feasible training strategy is to use a strong LLM as a teacher to generate structured relevance judgments and explanatory rationales for large numbers of

query–document pairs. These weak labels can then be reviewed selectively and distilled into a smaller Cross-Encoder. Hard-negative mining should be included so that the model learns to reject superficially related literature. Human annotation remains necessary for calibration and final evaluation, but LLM-assisted labeling may significantly reduce the burden of creating the initial training corpus. The resulting model should be compared with embedding similarity, zero-shot LLM scoring, and conventional Cross-Encoder baselines to determine whether domain-aware distillation produces a measurable improvement.

The information-flow design is another important contribution because it separates perception, retrieval, evidence selection, and language generation. The Context Pack prevents the final LLM from receiving an uncontrolled mixture of predictions and documents. Instead, the original query, species hypothesis, visual description, selected evidence, source metadata, and response constraints remain explicitly represented. This improves modularity and creates the possibility of tracing an incorrect answer back to a specific stage.

However, the present evaluation is not sufficient to prove that every information-flow decision is optimal. Future experiments should test whether including image captions improves retrieval, whether species confidence should alter query expansion, whether uncertain visual predictions should be represented as multiple hypotheses, and whether natural-history sources improve or bias the final answer. The framework should also measure evidence coverage, citation correctness, unsupported-claim rate, answer consistency, and uncertainty calibration. These evaluations would transform information-flow design from an architectural intuition into an empirically validated contribution.

At the complete-framework level, the preliminary average score of approximately 5.7 out of 10 indicates that *E pluribus* is not yet a strong ecological research

assistant. Its performance remains inconsistent across species identification, habitat analysis, behavioral interpretation, health-related questions, and research planning. Nevertheless, the system has achieved a meaningful transition beyond disconnected model demonstrations. It can process an observation, construct a species hypothesis, retrieve ecological literature, filter evidence, and produce a research-oriented response. This end-to-end behavior confirms that the project is aligned with the intended concept of ecological assistance, even though individual modules remain immature.

Future evaluation should employ systematic ablation studies. Each experiment should remove or replace one component, including image context, taxonomic expansion, hybrid retrieval, source-quality scoring, Cross-Encoder reranking, natural-history knowledge, or structured Context Pack rules. The resulting changes in retrieval relevance, evidence grounding, and response quality would reveal which components genuinely contribute to the final outcome. Evaluation should also involve multiple images, species, habitats, and question types rather than relying on a single representative observation. Ecologists or biology students could provide expert relevance judgments, while LLM-based evaluation could be retained as a scalable supplementary measure rather than the sole evaluator.

The broader software platform should also extend beyond a single analysis interface. Future versions may include searchable observation histories, document libraries, comparison of multiple field images, interactive taxonomic trees, geographic and temporal filters, exportable evidence reports, researcher annotations, citation management, and local project workspaces. Module-level confidence, retrieval provenance, and uncertainty should be visible to the user rather than hidden behind a final answer. Such features would move E pluribus from a technical framework toward a usable research environment.

Overall, the project contributes an integrated perspective on ecological artificial intelligence. Its principal finding is that effective research assistance requires more than accurate perception or fluent generation. It requires careful coordination among visual representation, domain-aware retrieval, contextual evidence selection, information provenance, and controlled synthesis. E pluribus remains an early-stage framework, but its experiments identify both the potential and the limitations of this direction. The next stage should be less concerned with preserving every original architectural choice and more concerned with selecting the most effective contemporary models, validating each information transition, and building a scientifically credible tool around the needs of ecological researchers.

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